# General Mathematics — Exercise Sheet 21

*Determinants and invertibility*

**Variant: Statements**

**Exercise 1 (Two rows changed at once).** A real $3 \times 3$ matrix $A$ has rows $r_1, r_2, r_3$ and determinant 5. For $c \in \mathbb{R}$, let $B_c$ have rows $r_1 + 2r_2$, $r_1 + cr_2$ and $-3r_3$, in that order. Compute $\det B_c$ using row linearity and the vanishing of repeated-row terms. Find all $c$ for which $B_c$ is singular. Keep the other rows fixed at each application of linearity.

**Exercise 2 (Three exceptional parameters).** For $t \in \mathbb{R}$, let

$$A_t = \begin{pmatrix} t & 1 & 0 \\ 1 & t & 1 \\ 0 & 1 & t \end{pmatrix}.$$

Use elimination with determinant factors recorded to compute $\det A_t$ for every $t$. Classify invertibility. For each singular parameter, give a basis of $\ker A_t$ and its dimension under Lecture 18's convention. Avoid dividing by $t$, since $t = 0$ is included.

**Exercise 3 (What determinant data determine).** Real $3 \times 3$ matrices $A$ and $B$ satisfy $\det A = -2$ and $\det B = 3$. Compute $\det(2A^{-1}B^2)$ and $\det(B^{-1}AB)$, explaining why both inverses exist and how many scalar factors the whole-matrix scaling contributes. To test whether those two determinant data also determine $\det(A + B)$, compare the concrete matrices

$$A_0 = \operatorname{diag}(-2, 1, 1), \quad B_1 = \operatorname{diag}(3, 1, 1), \quad B_2 = \operatorname{diag}(1, 3, 1).$$

Here diag denotes the diagonal matrix with the listed diagonal entries and all other entries zero. Compute $\det(A_0 + B_1)$ and $\det(A_0 + B_2)$ and state your conclusion.

**Exercise 4 (A zero diagonal entry before elimination).** Compute the determinant of the real matrix

$$D = \begin{pmatrix} 1 & 2 & 0 & 3 & 1 \\ 2 & 5 & 0 & 6 & 2 \\ 0 & 0 & 0 & 2 & 1 \\ 0 & 0 & 3 & 1 & -1 \\ 0 & 0 & 6 & 0 & 1 \end{pmatrix}$$

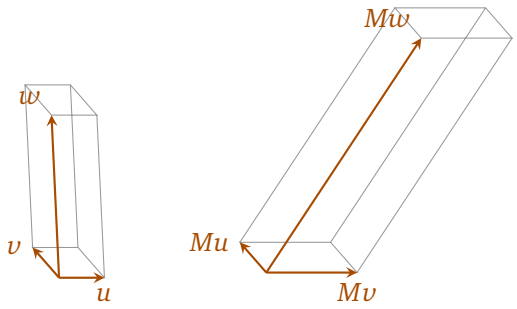
by Gaussian elimination, recording swaps and scalings. Explain why the initial zero in position $(3, 3)$ does not itself prove that the matrix is singular. State whether the resulting system $Dx = b$ has a unique solution for every $b \in \mathbb{R}^5$.

**Exercise 5 (Signed volumes under a real map).** Let a real linear map have matrix

$$M = \begin{pmatrix} 0 & 2 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 2 \end{pmatrix}.$$

An ordered parallelepiped has side columns $u = (1,0,0)$, $v = (0,1,0)$ and $w = (1,2,3)$. Find its signed and ordinary volumes, then find the image columns $Mu$, $Mv$ and $Mw$. Use determinant multiplicativity to find the image's signed and ordinary volumes and determine whether the orientation reverses. Find the signed volume obtained by applying $M^{-1}$ to the original three columns, and the signed volume obtained by applying $M$ twice to those original columns. Use Lecture 13's scalar-triple-product meaning and Lecture 21's product and inverse laws.

The drawings use the same projection and coordinate scales for the original and image parallelepipeds.




**Exercise 6 (A Vandermonde determinant and three prescribed values).** For $a, b, c \in \mathbb{R}$, compute the determinant

$$V = \begin{pmatrix} 1 & a & a^2 \\ 1 & b & b^2 \\ 1 & c & c^2 \end{pmatrix}$$

by row reduction and factoring row scalars. Obtain a formula valid also when two nodes coincide, without dividing by any difference of nodes. Determine exactly when a polynomial $q(x) = \alpha + \beta x + \gamma x^2$ of degree at most 2 is uniquely determined by arbitrary prescribed values $q(a) = y_a$, $q(b) = y_b$ and $q(c) = y_c$. Use Lecture 20's all-right-sides inversion criterion.